Ogbonna Arinze

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SUMMARY

Frontend Infrastructure Engineer with 8 years of professional experience building, optimizing, and scaling frontend platforms across healthcare, fintech, and civic tech. Specialized in monorepo architecture (Nx, pnpm workspaces), CI/CD pipeline design (GitHub Actions, Docker), build tooling optimization (Webpack, Rollup), and automated testing frameworks (Jest, Playwright). Proven track record of improving developer experience through engineering standards, performance profiling, bundle analysis, and reusable component systems, enabling stream-aligned teams to ship faster with confidence.

CORE COMPETENCIES

- **Frontend Infrastructure:** Monorepo orchestration (Nx, pnpm), micro-frontend architecture, CI/CD pipelines, build optimization, and developer tooling.
- **Build Tools & Bundlers:** Webpack, Rollup, Vite, Babel, webpack-bundle-analyzer, rollup-plugin-visualizer, and code splitting strategies.
- **Testing & Quality:** Jest, Enzyme, Playwright, Cypress, Allure Reports, Storybook, ESLint, Prettier, Husky, Lint-staged, and CI-driven QA.
- **Frameworks & Libraries:** React, Next.js, TypeScript, Vue.js, Redux, Zustand, React Query, Tailwind CSS, Radix UI, and Ant Design.
- **Cloud & DevOps:** GitHub Actions, Docker/Docker Compose, AWS (CloudFront, S3), Vercel, Traefik, multi-stage Dockerfiles, and health checks.
- **Performance & Observability:** Lighthouse, React Profiler, Chrome DevTools, Core Web Vitals, bundle analysis, CDN strategies, Sentry, and Datadog RUM.
- **Programming Languages:** TypeScript, modern JavaScript (ESNext), NodeJS, HTML, CSS, and CSS-in-JS.
- **Leadership:** Technical architecture decisions, engineering standards, mentorship, code reviews, Agile/Scrum, and cross-team developer experience advocacy.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, managing the parallel maintenance of the legacy codebase while shipping the new architecture.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, identifying and resolving bottlenecks using **Lighthouse** and **Chrome DevTools** to reduce load times significantly.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components, streamlining data flow and improving the developer experience across the codebase.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **GitHub Actions CI/CD pipeline**, automating test reporting with **Allure Reports** on GitHub Pages and configuring **Slack notifications** for build quality visibility on every push.
- Implemented **JWT-based authentication** and **role-based access control** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using OpenAI GPT models, **LangChain**, Vercel AI SDK, and Chroma vector database, enabling conversational data querying.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality through structured code reviews.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing **bundle size by 30-40%** through **rollup-plugin-visualizer** and **webpack-bundle-analyzer**, achieving a **35% faster load time** and smoother runtime experience across devices.
- Established engineering standards for code quality, testing, and deployment across the team using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating **CI/CD** processes.
- Enhanced the development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and nurturing the team toward test automation best practices.

- Drove a **40% improvement** in React application performance by optimizing caching mechanisms, significantly boosting speed and user experience.
- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Utilized **Storybook** for documenting React components, streamlining change management and ensuring clear communication of component APIs and usage patterns across development teams.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code and consistent design across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**, developing critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React, Redux, RxJS**, and **Axios**.
- Adopted the native browser **EventSource API** for real-time data streaming, replacing conventional socket.io methods for faster data updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design** UI.
- Developed a real-time account reconciliation solution, boosting the reconciliation of daily transactions by **60%** through innovative transaction processing and data management strategies.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, and devised comprehensive documentation for each application detailing functions, capabilities, and features.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications** with **pnpm workspaces**: NestJS API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** end-to-end tests across 4 test projects (web auth, dashboard auth, desktop Chrome, mobile Pixel 5), and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown.
- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts** (22 interactive chart types), featuring dark mode, URL state management, responsive mobile layouts, and **SSR** with SEO optimization (Open Graph, JSON-LD, slug-based URLs).
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, wrote custom migration tooling, and integrated **Traefik** for reverse proxy routing, **Langfuse** for observability, and **Infisical** for secrets management across environments.
- Built a multi-agent AI system using the **Mastra** framework with 5 specialized agents, a hybrid **RAG pipeline** combining pgvector + BM25 full-text search with Reciprocal Rank Fusion, and **SSE streaming** with 15s keepalive heartbeats.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*